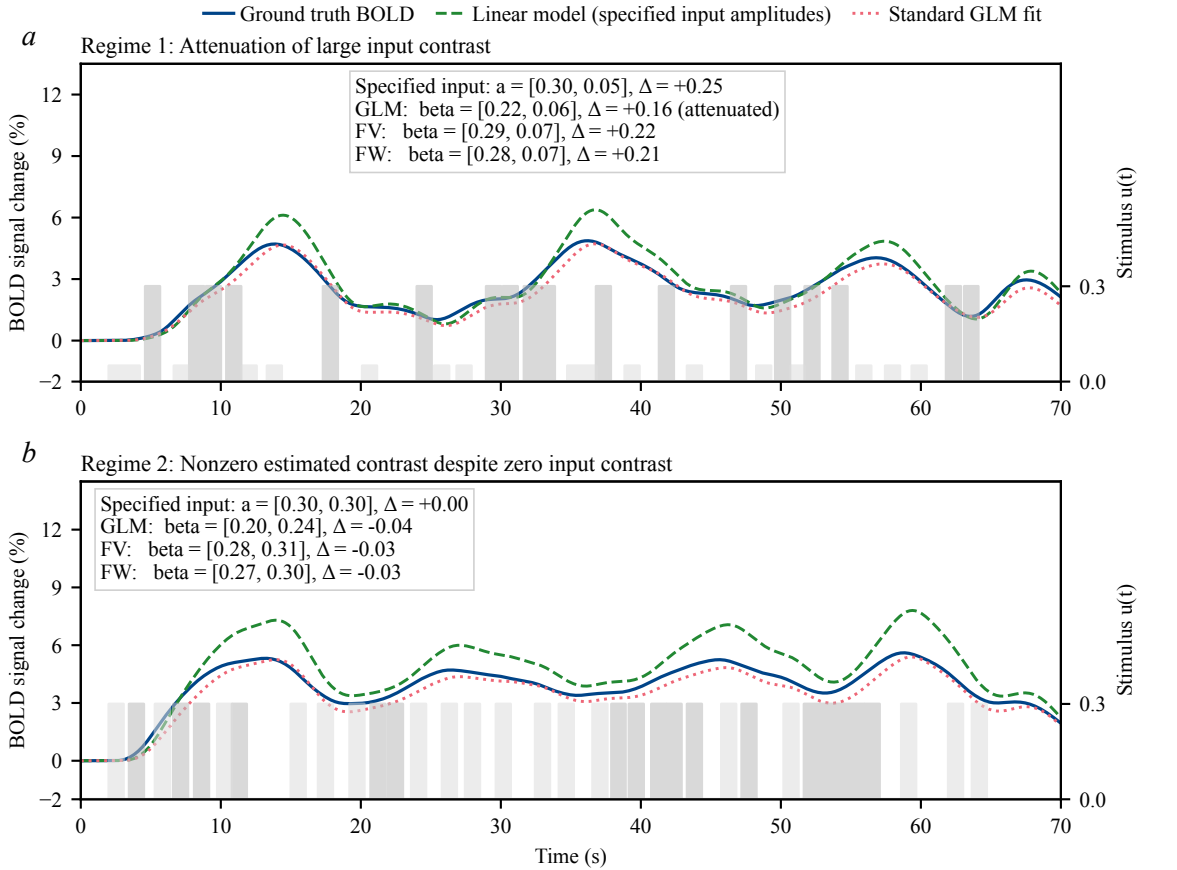

Ground truth BOLD
Linear model (specified input amplitudes)
Standard GLM fit
a
Regime 1: Attenuation of large input contrast
Specified input: a = [0.30, 0.05], Δ = +0.25
GLM: beta = [0.22, 0.06], Δ = +0.16 (attenuated)
FV: beta = [0.29, 0.07], Δ = +0.22
FW: beta = [0.28, 0.07], Δ = +0.21
BOLD signal change (%)
Stimulus u(t)
b
Regime 2: Nonzero estimated contrast despite zero input contrast
Specified input: a = [0.30, 0.30], Δ = +0.00
GLM: beta = [0.20, 0.24], Δ = -0.04
FV: beta = [0.28, 0.31], Δ = -0.03
FW: beta = [0.27, 0.30], Δ = -0.03
BOLD signal change (%)
Stimulus u(t)
Time (s)